Input system:
Dependent variables:x
Independent variables:t
Equations:

$$\frac{dx}{dt} = ax\left(x - L\right)\mathrm{inv}\left(N + x\right)$$

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Result:
Dependent variables:s_1, x
Independent variables:t
Additional variables:

$$s_1 = \mathrm{inv}\left(N + x\right)$$

Equations:

$$\frac{dx}{dt} = s_1 a x^2 + \left(-1\right) s_1 a x L$$

$$\frac{ds_1}{dt} = \left(-1\right) s_1^3 a x^2 + s_1^3 a x L$$